

Tester(QA) Career Research & Learning Plan

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1. Target Role

Role: Tester (Quality Assurance, Test Analyst)

2. Research the Market

- **Key Responsibilities:**
 - Developing, reviewing, and executing test cases/scripts.
 - Performing functional, integration, regression, and UAT testing.
 - Identifying, documenting, and tracking defects using bug-tracking software.
 - Validating data integrity and performing SQL queries.
 - Collaborating with developers, BAs, and stakeholders in Agile/Scrum teams.
- **Required Skills:**
 - Strong analytical and troubleshooting mindset with meticulous attention to detail.
 - Good understanding of SDLC and testing methodologies.
 - Basic programming/scripting (Python, C++, or C#) and SQL database knowledge.
 - Strong written and verbal communication skills.
- **Tools and Technologies:**
 - **Defect & Test Management:** Jira, Confluence, OpenText ALM, Azure DevOps.
 - **Database & Query:** SQL Server, SQL Management Studio (SSMS).
 - **Automation & Scripting:** Python (Pytest), Selenium (optional/desirable).
 - **Collaboration:** Microsoft Office Suite, Teams, SharePoint.
- **Certifications Commonly Requested:**
 - **ISTQB Foundation Level** (International Software Testing Qualifications Board)

3. Create a Skills & Resources Table

Tool/Technology	Purpose	Learning Resource
Jira / Confluence	Defect tracking, test case management, and agile collaboration	LinkedIn Learning, Atlassian University
SQL / SSMS	Data validation, backend verification, and database querying	LinkedIn Learning, YouTube (FreeCodeCamp)
Python / Pytest	Writing automated tests and scripting for embedded/software systems	LinkedIn Learning, Python.org tutorials
ISTQB Concepts	Understanding testing principles, lifecycle, and design techniques	ASTQB syllabus, Udemy, YouTube

4-Week Study Plan for Tester (QA)

Week	Focus Area	Recommended Courses	Practical Exercise & Deliverables
Week 1	Manual Testing Foundations	• Software Testing Foundations: Fundamentals • Software Testing Foundations: Test Techniques • Agile Testing	• Pick any public website/app. • Write a test plan and 15 manual test cases (functional, negative, boundary) for a feature (e.g., login form). • Log 3 intentional bugs into a free tracker (e.g., Trello).
Week 2	Automation Basics & API	• Selenium Essential Training • Learning Test Automation • API Testing Foundations	• Automate 5 manual test cases from Week 1 using Selenium (Python). • Use Postman to write 5 API test requests (GET/POST) against a public API (e.g., reqres.in), asserting status codes and response fields.
Week 3	CI/CD & Test Management	• Jenkins for Developers/Testers (or Learning DevOps: CI/CD) • Jira Essential Training • Software Testing: Bug Reports and Test Cases	• Push Selenium tests to a GitHub repository. • Set up a free GitHub Actions workflow to run tests automatically on push. • Log all bugs found across weeks into a Jira board with severity/priority tags.
Week 4	Advanced Practice & Portfolio	• Advanced QA Testing & Interview Prep • SQL for Testers (Data Validation) • Portfolio Building & Resume Optimization	• Perform backend data verification using SQL queries on a sample database. • Document an end-to-end test summary report. • Polish your QA portfolio and practice common testing interview scenarios.